

Suite Hard · Simulation Module 01

22 questions · 35 minutes · official SAT Math Hard

Exclude-active: these items are not on current Bluebook full-length tests.

All items are Hard. This is not an adaptive Module 1. It is harder than test day.

Calculator / Desmos is allowed on every item, same as Bluebook.

Question 1 is the first page after this cover. Write answers on the last page.

Read the prompt carefully. Write the job on scratch before you copy numbers.

Domain mix in this module:

Algebra: 5

Advanced Math: 8

PSDA: 4

Geometry: 5

Do not open the next module until this one is timed and scored.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

Store A sells raspberries for **\$5.50** per pint and blackberries for **\$3.00** per pint. Store B sells raspberries for **\$6.50** per pint and blackberries for **\$8.00** per pint. A certain purchase of raspberries and blackberries would cost **\$37.00** at Store A or **\$66.00** at Store B. How many pints of blackberries are in this purchase?

Answer

- A. 4
- B. 5
- C. 8
- D. 12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

$a, 26, 29, b, 31, 47, c$

For the given data set, the data values are listed in ascending order, where a , b , and c are constants. For this data set, the mean is **36**, the median is **29**, and the range is **72**. What is the value of c ?

Answer

- A. **54**
- B. **72**
- C. **81**
- D. **98**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$h(x) = x^3 + ax^2 + bx + c$

The function h is defined above, where a, b, and c are integer constants. If the zeros of the function are -5 , 6, and 7, what is the value of c ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

The given equation, where t is a constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{83}$. What is the value of t ?

Answer

- A. 73
- B. 71
- C. 37
- D. 35

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

Adam’s school is a 20-minute walk or a 5-minute bus ride away from his house. The bus runs once every 30 minutes, and the number of minutes, w , that Adam waits for the bus varies between 0 and 30. Which of the following inequalities gives the values of w for which it would be faster for Adam to walk to school?

Answer

- A. $w - 5 < 20$
- B. $w - 5 > 20$
- C. $w + 5 < 20$
- D. $w + 5 > 20$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

	Blood type			
Rhesus factor	A	B	AB	O
+	33	9	3	37
−	7	2	1	x

Human blood can be classified into four common blood types—A, B, AB, and O. It is also characterized by the presence (+) or absence (−) of the rhesus factor. The table above shows the distribution of blood type and rhesus factor for a group of people. If one of these people who is rhesus negative (−) is chosen at random, the probability that the person has blood type B is $\frac{1}{9}$. What is the value of x ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which of the following is equivalent to $\left(a + \frac{b}{2}\right)^2$?

Answer

A. $a^2 + \frac{b^2}{2}$

B. $a^2 + \frac{b^2}{4}$

C. $a^2 + \frac{ab}{2} + \frac{b^2}{2}$

D. $a^2 + ab + \frac{b^2}{4}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy-plane, the graph of $2x^2 - 6x + 2y^2 + 2y = 45$ is a circle. What is the radius of the circle?

Answer

- A. 5
- B. 6.5
- C. $\sqrt{40}$
- D. $\sqrt{50}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

In the xy -plane, line k intersects the y -axis at the point $(0, -6)$ and passes through the point $(2, 2)$. If the point $(20, w)$ lies on line k , what is the value of w ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

0.1152 is 0.08% of the number b . The number b is 25% of the number c . What is the value of c ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

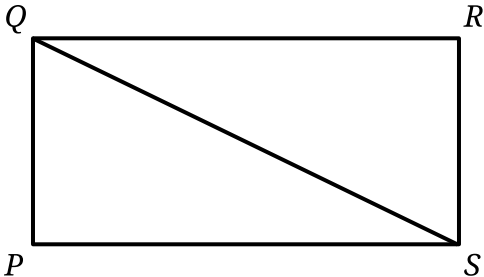
The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$. What is the value of ab ?

Answer

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 54
- D. 60

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS . What is the length of side QR ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$\frac{12x+28}{4} - \frac{s}{13} = r(x-8)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

A reseller buys certain books for a purchase price of **5.00** dollars each and then marks them each for sale at a consumer price that is **270%** of the purchase price. After **4** months, any remaining books not yet sold are marked at a discounted price that is **70%** off the consumer price. What is the discounted price of each of the remaining books, in dollars?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A model estimates that at the end of each year from **2015** to **2020**, the number of squirrels in a population was **150%** more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of **2016**, there were **180** squirrels in the population. Which of the following equations represents this model, where ***n*** is the estimated number of squirrels in the population ***t*** years after the end of **2015** and ***t* ≤ 5**?

Answer

- A. ***n* = 72(1.5)^{*t*}**
- B. ***n* = 72(2.5)^{*t*}**
- C. ***n* = 180(1.5)^{*t*}**
- D. ***n* = 180(2.5)^{*t*}**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

Answer

- A. $-14x + 1,540$
- B. $14x - 320$
- C. $-28x + 1,720$
- D. 360

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

A team hosting an event to raise money for new uniforms plans to sell at least **140** tickets before this event and at least **220** tickets during this event to raise a total of at least **\$5,820** from all tickets sold. The price of a ticket during this event is **\$3** less than the price of a ticket before this event. Which inequality represents this situation, where x is the price, in dollars, of a ticket sold during this event?

Answer

- A. $140(x + 3) + 220x \leq 5,820$
- B. $140(x + 3) + 220x \geq 5,820$
- C. $140(x - 3) + 220x \leq 5,820$
- D. $140(x - 3) + 220x \geq 5,820$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$y = x^2 + 2x + 1$
 $x + y + 1 = 0$
If (x_1, y_1) and (x_2, y_2) are the two solutions to the system of equations above, what is the value of $y_1 + y_2$?

Answer

- A. -3
- B. -2
- C. -1
- D. 1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = 4x^2 - 50x + 126$

The given equation defines the function f . For what value of x does $f(x)$ reach its minimum?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

x	y
21	−8
23	8
25	−8

The table shows three values of x and their corresponding values of y , where $y = f(x) + 4$ and f is a quadratic function. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. What is the value of r ?

Module 01 • answer sheet

Write one answer per line. SPR: integer, decimal, or fraction.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

15. _____

16. _____

17. _____

18. _____

19. _____

20. _____

21. _____

22. _____